

Probabilistic Analysis Enabled Proactive Fault Risk Assessment of Electricity Power Distribution Systems

Haisheng Hong*

Guangzhou Power Supply Bureau of
Guangdong Power Grid Co., Ltd
Guangzhou, China
honeyhycere@qq.com

*Corresponding author

Zhixuan Zhao

Guangzhou Power Supply Bureau of
Guangdong Power Grid Co., Ltd
Guangzhou, China
1285752210@qq.com

Zhifang Zhu

Guangzhou Power Supply Bureau of
Guangdong Power Grid Co., Ltd
Guangzhou, China
13826432504@163.com

Zheng Sun

Guangzhou Power Supply Bureau of
Guangdong Power Grid Co., Ltd
Guangzhou, China
sunzheng_gzj@126.com

Shengkai Chen

Guangzhou Power Supply Bureau of
Guangdong Power Grid Co., Ltd
Guangzhou, China
849482437@qq.com

Zihao Weng

Guangzhou Power Supply Bureau of
Guangdong Power Grid Co., Ltd
Guangzhou, China
401982574@qq.com

Abstract—This paper establishes a probabilistic operational scenario forecasting (POSF)-based method to obtain the actual operational state profile and probabilistic map for power distribution system reliability analysis (PFRA) tasks. The proposed POSF forecasts various probabilistic scenarios with multi-scale lead times based on Wasserstein Generative Adversarial Networks (WGAN), and the developed WGAN model is further incorporated into a stochastic constrained optimization framework to establish the conditional confidence interval and risk assessment index based on past observations. The POSF-based method is extensively evaluated through a range of power outage events in power distribution system transformers' daily operation realistic measurement records. The simulation outcomes confirm that the developed method can well adaptively forecast operational scenarios under uncertainties along with outage risk assessment compared with the existing benchmarks.

Keywords—risk assessment, electricity power distribution networks, fault analysis, probabilistic analysis

I. INTRODUCTION

The fundamental task of developing a low-carbon power system is to form a stable low-carbon power supply mechanism, and the key lies in the effective development and utilization of renewable energy. There are currently two basic ideas for this: one is to vigorously develop long-distance, high-capacity, and low loss cross regional transmission lines (e.g., ultra-high voltage transmission) to achieve optimized allocation of renewable energy resources in a larger regional scope. The other pathway is to start from the distribution and utilization of electricity, establish an integrated power supply system that coordinates and associates Distributed Renewable Energy Generation (DREG), distribution network, and terminal electricity, and achieve on-site consumption and utilization of renewable energy. Compared to the former, distributed power distribution systems have the advantages of short construction period, low investment cost, flexible operation,

and stronger risk resistance, thus gaining widespread attention worldwide in recent years.

The Proactive Fault Risk Assessment (PFRA) is expected to provide early warning of possible power outage risks based on the current system's observed operating states. This helps the operators to carry out timely corrective actions if necessary to guarantee the safety and reliability of the system operation. Since the dynamics of power distribution network operations may introduce more contingency conditions that can lead to an increase in operational reliability risks and even power outages and blackouts [1], the PFRA is of paramount importance to improve the system reliability and resilience for power distribution grids. Despite of capturing the stochastic behavior of the system in PFRA, few studies have focused on this aspect due to the challenges involved in characterizing uncertainties. These challenges include accurately assigning inputs to reproduce historical data patterns, correlating multidimensional operating parameters to approach the correct operating state series, and tuning prediction model parameters using an appropriate algorithm.

In practice, the PFRA for the power distribution system in the presence of uncertainties is considered a complex and challenging issue for several reasons: (1) the massive connection of different forms of renewable power generation resources has introduced rapidly changing operating conditions to the distribution network, which increases operational risk insecurity and the risk of power outages [2]; (2) In rapidly changing operating conditions and disorderly load fluctuations, risk features can be hardly observed from analysis of univariate data without considering correlated factors, which can lead to omissions and misstatements in fault detection; and (3) there is a lack of quantitative indicators for risk assessment in achieving the best trade-offs for warning performance given the inaccuracy of measurements and the lack of coverage [3].

To address the aforementioned issues and support the daily operation of the power distribution network, short-term probabilistic operation state forecasting is considered

suitable for day-ahead observations in PFRA applications. Since day-ahead point forecasting may significantly deviate from the actual operational state due to uncertainties. The interval prediction with an operation interval can be adopted for flexible analysis in a single plot, and can capture the stochastic uncertainties for prediction. Based on the studies in [4] and [5], the prediction uncertainties can be observed, and hence to promote the system operational risk assessment and decision-making. However, the interval prediction solution based on statistical mean-variance from a predictor's probability density function may not provide the clarity required for analysis of disorderly operation states.

In parallel, probabilistic forecasting provides a probabilistic distribution with uncertain information for chance constraints. The majority of studies on probabilistic forecasting are statistically-based methods. The authors in [6] proposed a nonparametric probabilistic electricity load forecasting approach and quantified the uncertainty with information entropy. The deep Gaussian processes-based probabilistic demand prediction solution was developed in [7] to identify the anomaly events. However, the available probabilistic forecasting methods build the predictor's probability density function based on statistical assumptions (e.g., Gaussian distribution), which is simple and convenient but lacks intended risk information on external uncertainties.

In addition, a set of machine learning-based non-parametric solutions for risk assessment has been explored. The work in [8] assessed the operational risk levels based on entropy-based fusion of particle filter prognostics. The authors in [9] proposed an entropy-based prognostics model for system state assessments based on scenarios generated by particle filters at different times. The study in [10] adopted the rare events logistic regression method to assess uncertainties of renewable energy generation in the power grids. However, in the case that the operational state is non-stationary, these methods need to be further extended to consider the probability distribution of the preferred observation periods. When the realistic operational state does not obey the pre-assumed distribution, the solutions cannot match the underlying mapping function [11]. The work in [12] exploited a spatio-temporal estimation generative adversarial network for high-resolution generation of measurements for improving the state estimation performance for power distribution systems. In addition, most recently [24] developed operational scenario prediction solution for integrated energy systems considering different forms of operational uncertainties.

In summary, the existing solutions for PFRA have the following limitations: (1) reliance on the Gaussian assumption in probabilistic operational forecasting, which can lead to a loss of sharp features in realistic distributions, especially for disorderly fluctuating renewable scenarios; (2) lack of temporal adaptability and rely on fixed distributions, and they do not consider the conditional probability in different scenarios; and (3) few solutions are available to present quantitative risk indicators to support the trade-off between the earliness and accuracy of risk assessment.

To this end, this paper proposes a probabilistic operational scenarios forecasting (POSF) based solution for the PFRA task. An WGAN-based network model is designed for unsupervised representation of the statistical properties inherent in historical operational data. The various scenario sets achieved based on a well-trained WGAN will be

integrated into a stochastic constrained optimization to adaptively construct operational state forecasting and compute the most probable quantile occurring in the training horizon. Through the analysis of the inputs (i.e., past observations and point forecasting), the confidence interval (CI) of different certainty levels is drawn, and the quantitative risk indicators are calculated as the reference for the decision on earliness and accuracy. The effectiveness of the solution is validated through extensive experiments based on real historical data and outage event sets.

The main contributions of this paper are as follows: The proposed POSF-based solution enables the actual portrayal of the operational state profile with probable quantile and risk indicator for advanced operational risk assessment in the power distribution network. The WGAN is designed to adaptively construct an unsupervised representation of inherent statistical properties of uncontrollable loads and generate realistic scenarios and probabilistic forecasting based on past observations.

The rest of this work is presented as follows: Section II presents the problem formulation and the proposed solution. The experiments are carried out and numerical results are provided in Section III. Finally, the conclusive remarks are given in Section IV.

II. PROBLEM FORMULATION AND PROPOSED SOLUTION

This section presents three tasks essential for capturing uncertainties in modeling uncertainties for power distribution network operation: (1) scenario generation, which involves producing a range of realistic operational scenarios; (2) scenario forecasting, which involves generating accurate operational scenarios for a specified upcoming period (such as a few hours) by incorporating relative certain information; and (3) risk assessment, which is a challenging task for judgment interval is ambiguous, and weak risk feature close to decision boundary may result in misclassification.

A. Probabilistic Operational Scenario Generation

The probabilistic operational scenario generation is the foundational task for scenario forecasting. The distribution network operation encompasses multiple scenarios that enable the uncertainty assessments. Each scenario is developed based on the distinct probabilities of essential predictors that affect the load forecasting performance. This study assumes that uncertainty arises from related observable information. The input variables $x \in X$ comprises essential factors that can affect the actual operational state, the X is the scenario set. For a typical site in a distribution network with disorderly load fluctuations, x represents the operational state vector. The features are selected by variables that cannot be planned and available from power monitoring systems, as shown in (1),

$$x = [I_a, I_b, I_c, U_a, U_b, U_c, L_{rate}, P, Q]^T \quad (1)$$

Where I_a , I_b and I_c represent three phases of currents of main circuits, U_a , U_b and U_c represent three phases of voltages of buses, L_{rate} represents the load rate of the distribution transformer, P and Q represent the active power and reactive power, respectively. Assuming at time step t , we have records for past h steps operation state

$x_{hist} = [x_{t-h} \dots x_t]^T, x \in X^h$. Meanwhile, denote the point forecasts $x_{pred} = [x_{t+1} \dots x_{t+k}]^T, x \in X^k$ for each look-ahead time $i = 1, \dots, k$ obtained by x_{hist} , where k is the forecasting step. Consider a general discrete-time state evolution system described as shown in (2), which models in quantiles to describe the distribution of the X as.

$$X = f(x_{hist}) + \varepsilon \quad (2)$$

Where x represents the state vector at each discrete time instant, $f(\bullet)$ is the system state forecasting function and ε is uncertainty residual. The forecasting function is run several times, where ε is the q th quantile of the residual. By using (3), a residual modeling is constructed in a quantile regression model as follows:

$$\varepsilon_q = F_q(\beta_q, [X, f(x_{hist})]) \quad (3)$$

Where F_q and β_q denote the q th quantile regression model and its corresponding parameters, respectively. The scenario generation X can be formulated as $S = \{S_1, \dots, S_n\}$ are generated to represent the possible variations around the point forecast and accurately reflect the dynamic fluctuations of operation state. S is an N -dimensional unity space that contains n scenarios. If $x \in S, x = (x_1, x_2, \dots, x_{h+k})$, then the probability of a set of x can be denoted as $\{P_n \subseteq P(S_{nL}), 0 \leq L \leq h+k\}$, which is as shown in (4), and the generation of scenarios is denoted as (5),

$$P(S) = \{P(x_1^{(L)}), \dots, P(x_i^{(L)}), \dots, P(x_n^{(L)}), 1 \leq i \leq n\} \quad (4)$$

$$S_{Lon} = G(z_{Lon}) \quad (5)$$

Where the $G(\bullet)$ is the function of scenario generation and the z is a random noise without regarding to x_{pred} and x_{hist} , $G(z)$ is the generated scenario, the n is the number of scenarios and the L is the length of time series. The scenarios model is to learn the distribution of $P(S)$ without any statistical assumptions.

B. Probabilistic Operational Scenarios Forecasting

The probabilistic forecasting for latent operational scenarios is obtained by optimization problem based on a maximum a posteriori scenarios generation issue. To determine the quality of the generated distribution, the discriminator $D(\bullet)$ is trained to score the $G(z)$, denoted as $D(G(z))$, and a high $D(\bullet)$ indicates that θ of the model $G(\bullet)$ is appropriate. The generated scenarios $G(z)$ need to satisfy two conditions: 1) the part of $G(z)$ from time step $t-h$ to t should be closed to the historical data x_{hist} ; and 2) the part of $G(z)$ from time step t to $t+k$ should be diverse and realistic regarding the point forecast. The scenario forecasting optimization can be defined as follows:

$$\text{Min}_z \|G(z)_{hist} - x_{hist}\|_2 - \gamma \|D(G(z))\|_2 \quad (6)$$

$$s.t. \quad z \in \mathbb{Z} \quad (7)$$

$$L^\xi(x_{pred}) \leq G(z) \leq U^\xi(x_{pred}) \quad (8)$$

$$[L^\xi(x_{pred}), U^\xi(x_{pred})] = \left[\frac{1}{\xi} x_{pred}, \xi x_{pred} \right] \quad (9)$$

Where the $\|G(z)_{hist} - x_{hist}\|_2$ is the distance of the true state and the generated scenario space, and the γ is a weighting parameter. The $L^\xi(\bullet)$ and $U^\xi(\bullet)$, controlled by the factor ξ , represent the low and high bound of the generator, respectively. To ensure that the generated scenarios do not conflict with the information provided by point forecasts, (9) uses the point forecast x_{pred} by defining a prediction interval.

After addressing a group of various scenario forecasting and the weighted coefficient of each sequence, the final fusion operational scenario $\tilde{X} = \{\tilde{x}_i | i = 1, 2, \dots, s\}$ is obtained as shown in (10) and (11).

$$P(\tilde{x}_i) = \sum \delta S - \sum_{i=1}^s \delta^i w^i x_i^j \quad (10)$$

$$\tilde{x} = \sum_{i=1}^n w^i x_i^j, \quad i = 1, 2, \dots, n, \quad \sum_{i=1}^n w^i = 1 \quad (11)$$

C. Risk Assessment

The fault risk assessment is considered a binary classification problem, as illustrated in Fig. 1. The idea behind this is to train the model to fit fluctuation correlations and distribution of the training dataset, and predict the regression outputs and make the classification based on the discriminator.

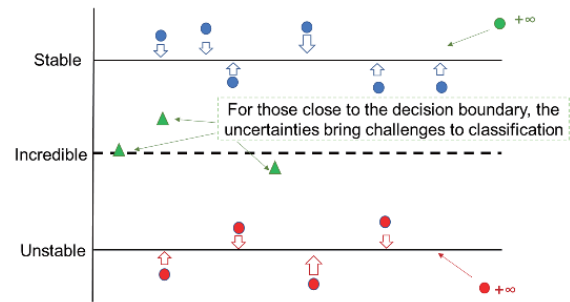


Fig 1. Fuzzy judgment interval for risk assessment, the incredible value may lead to false alarms

The judgment interval for risk assessment is ambiguous, and incorrect values can directly lead to false alarms based on the classification model. In practice, the risk event in the early period with the weak feature may have regression outputs that are very close to the decision boundary.

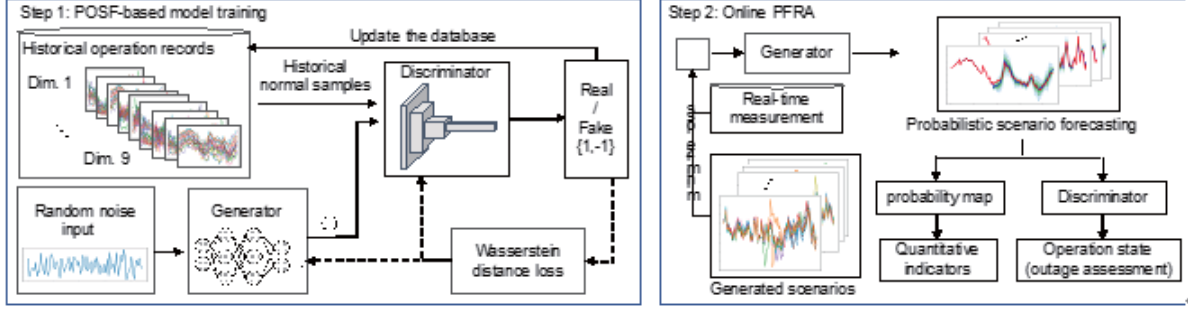


Fig 2. Overview of the framework of POSF-based PFRA task

$$\text{if } \begin{cases} D(G(z)) \in [lb_s, ub_s] & (\text{stable}) \\ D(G(z)) \in [lb_u, ub_u] & (\text{unstable}) \\ D(G(z)) \in (ub_u, lb_s) \forall (-\infty, lb_s) \forall (ub_s, +\infty) & (\text{incredible}) \end{cases} \quad (12)$$

Where the $D(G(z))$ is the score of classifier, $[lb_s, ub_s]$, $[lb_u, ub_u]$ and $(ub_u, lb_s) \forall (-\infty, lb_s) \forall (ub_s, +\infty)$ are the boundaries to define the stable, unstable and incredible intervals of $G(z)$. To quantitatively evaluate the performance of the risk assessment, two indices, credibility $p_c(x)$ and accuracy $p_a(x)$, are defined as (13) and (14).

$$p_a(x) = \frac{n_{acc}}{n} \quad (13)$$

$$p_c(x) = P(z_{1-\alpha/2} \leq \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \leq z_{\alpha/2}) \quad (14)$$

Where n is the number of tested events, n_{acc} is the number of accurately classified event, the z is the distribution of scenarios and α is the confidence interval.

A novel advanced PFRA solution based on POSF is developed in Fig. 2, which contains the POSF-based model training and online advanced PFRA. In the former stage, a scenario generator based on a fluctuating measurement matrix is built to characterize the operational scenarios with great uncertainties. Based on the realistic probabilistic operational scenarios, the discriminator is trained as an operational risk assessment classifier with Wasserstein distance, and the instances are classified into stable cases (with values in $[lb_s, ub_s]$), unstable cases (with values in $[lb_u, ub_u]$) and incredible cases (with other conditions). The generator and the discriminator constantly loop to update one and fix the other, and hence adaptively combining probabilistic forecasting and risk assessment.

III. SIMULATION EXPERIMENTS AND EVALUATION

This section analyzes typical operation risk assessment cases for outage events of credibility and accuracy. The true and false alarms are marked with yellow and gray boxed in the measurement series, respectively. Fig. 3 and Fig. 4 show the red curves and blue curves with confidence intervals represent the actual values and estimated value, respectively.

Case 1: sub-switch tripping fault. Fig. 3 presents a case of sub-switch tripping fault with the comparison of probabilistic scenarios and actual values in various operation features. Fig. 3 demonstrates the event of branch circuit trip occurred at the 19th step of the measurement series (4:45 A.M.), and the POSF-based risk assessment is of 21 hours lead time (beginning from the 12th time step with yellow boxed). The true value has significant deviations from the estimated operational state, especially in features of currents. The assessment is accurate and timely with the current A sudden disappearance and the plunge of currents of three phases and the voltages of A phase is out of the confidence interval of stable operational state.

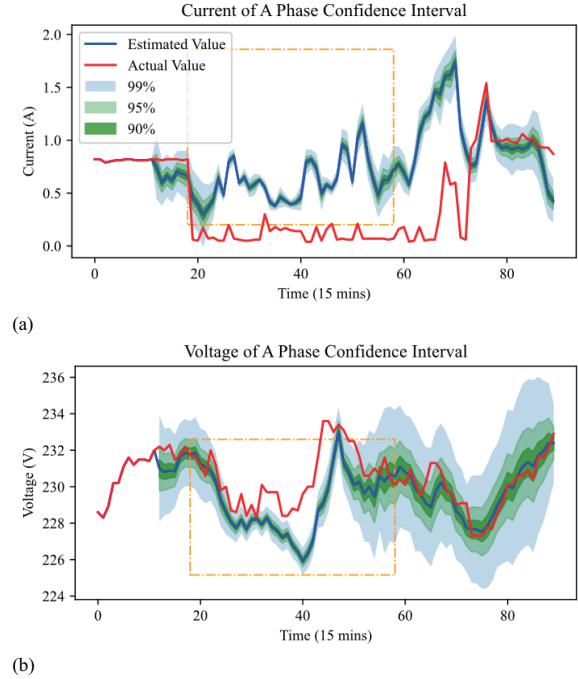


Fig 3. Operational scenarios for outage event for Case 1 with 21 hours prediction lead time. (a) Currents of Phase A; and (b) Voltages of Phase A.

Case 2: False alarms and real faults in region with disorderly load fluctuations. For region with disorderly load fluctuations, there are frequent dramatic fluctuations that often be misjudged as outage events, and the real risks are easily missed. Fig. 4 presents a daily operation state in a distribution network with highly volatile loads, which contains normal state with a sudden plunge at the 24th time step (in the grey box) and an outage event at the 66th time

step (in the yellow box). As shown in Fig. 4, although the current dropped to a nearly-zero level at the 24th time step, there was no outage and the confidence interval (CI) of POSF-based predicted voltage and current can cover the actual value, which proves the POSF-based method can avoid false alarms. At the same time, an outage happened at the 66th time step with currents and the active power plunges correctly identified since the actual value is incredible for the estimated value. It shows the POSF-based solution outperforms the benchmarks operational risk assessment task with abilities of probabilistic scenario forecasting under uncertainties.

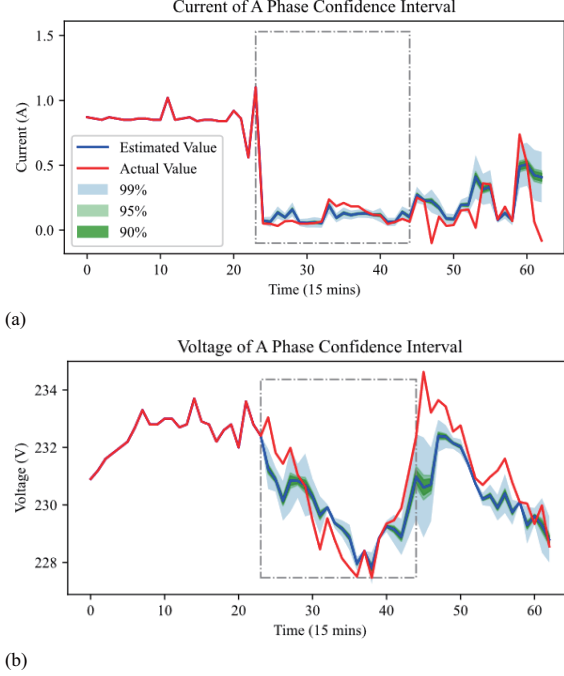


Fig 4. Operational scenarios for outage events for case without false alarm for normal operation state with sudden plunges: (a) current of Phase A; and (b) voltage of Phase A.

IV. CONCLUSION

This paper presents a risk assessment solution based on probabilistic operational scenarios forecasting for outage event detection for power distribution systems. In detail, a probabilistic operational scenario forecasting (POSF)-based method to obtain the actual operational state profile and probabilistic map for power distribution system reliability analysis (PFRA) tasks. The proposed method is validated through a set of simulation experiments and the simulation results confirmed its effectiveness and benefits.

For future work, the proposed method can be further improved and validated in practical deployment at the medium and low voltage power distribution networks. The model can be further improved based on the training with different datasets collected from the operation of power distribution systems. In addition, the risk analysis can also be compared with the conventional statistical solutions for a range of operational scenarios of the power distribution systems.

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AUTHOR CONTRIBUTIONS

The corresponding author is Haisheng Hong

REFERENCES

- [1] Gholami, Amin, Tohid Shekari, and Santiago Grijalva. "Proactive management of microgrids for resiliency enhancement: An adaptive robust approach." *IEEE Transactions on Sustainable Energy* 10.1 (2017): 470–480.
- [2] X. Shi, R. Qiu, Z. Ling, F. Yang, H. Yang, and X. He, "Spatio-Temporal Correlation Analysis of Online Monitoring Data for Anomaly Detection and Location in Distribution Networks," *IEEE Trans. Smart Grid*, vol. 11, no. 2, pp. 995–1006, Mar. 2020, doi: 10.1109/TSG.2019.2929219.
- [3] F. L. Grando, A. E. Lazzaretti, and M. Moreto, "The Impact of PMU Data Precision and Accuracy on Event Classification in Distribution Systems," *IEEE Trans. Smart Grid*, vol. 13, no. 2, pp. 1372–1382, Mar. 2022, doi: 10.1109/TSG.2021.3126268.
- [4] S. H. Begg, M. B. Welsh, and R. B. Bratvold, "Uncertainty vs. Variability: What's the Difference and Why is it Important?," in *Day 1 Mon, May 19, 2014, Houston, Texas: SPE, May 2014*, p. D011S003R002. doi: 10.2118/169850-MS.
- [5] S. M. Mazhari, N. Safari, C. Y. Chung, and I. Kamwa, "A Quantile Regression-Based Approach for Online Probabilistic Prediction of Unstable Groups of Coherent Generators in Power Systems," *IEEE Trans. Power Syst.*, vol. 34, no. 3, pp. 2240–2250, May 2019, doi: 10.1109/TPWRS.2018.2888831.
- [6] C. Wan, Z. Cao, W.-J. Lee, Y. Song, and P. Ju, "An Adaptive Ensemble Data Driven Approach for Nonparametric Probabilistic Forecasting of Electricity Load," *IEEE Trans. Smart Grid*, vol. 12, no. 6, pp. 5396–5408, Nov. 2021, doi: 10.1109/TSG.2021.3101672.
- [7] D. Cao et al., "Robust Deep Gaussian Process-Based Probabilistic Electrical Load Forecasting Against Anomalous Events," *IEEE Trans. Ind. Inform.*, vol. 18, no. 2, pp. 1142–1153, Feb. 2022, doi: 10.1109/TII.2021.3081531.
- [8] X. Shi, R. Qiu, T. Mi, X. He, and Y. Zhu, "Adversarial Feature Learning of Online Monitoring Data for Operational Risk Assessment in Distribution Networks," *IEEE Trans. Power Syst.*, vol. 35, no. 2, pp. 975–985, Mar. 2020, doi: 10.1109/TPWRS.2019.2941162.
- [9] Y. Wang, K. Jiang, and L. Xu, "Sensor-Data-Driven Fusion Prognostic Framework for Complex Engineered Systems," *IEEE Sens. J.*, vol. 21, no. 18, pp. 20421–20430, Sep. 2021, doi: 10.1109/JSEN.2021.3095556.
- [10] X. Chen, J. Qiu, L. Reedman, and Z. Y. Dong, "A Statistical Risk Assessment Framework for Distribution Network Resilience," *IEEE Trans. Power Syst.*, vol. 34, no. 6, pp. 4773–4783, Nov. 2019, doi: 10.1109/TPWRS.2019.2923454.
- [11] J. M. Morales, R. Míguez, and A. J. Conejo, "A methodology to generate statistically dependent wind speed scenarios," *Appl. Energy*, vol. 87, no. 3, pp. 843–855, Mar. 2010, doi: 10.1016/j.apenergy.2009.09.022.
- [12] Y. Liu, Y. Wang and Q. Yang, "Spatio-Temporal Generative Adversarial Network Based Power Distribution Network State Estimation With Multiple Time-Scale Measurements," in *IEEE Transactions on Industrial Informatics*, vol. 19, no. 9, pp. 9790–9797, Sept. 2023.
- [13] Y. Wang, Y. Liu and Q. Yang, "Operational Scenario Generation and Forecasting for Integrated Energy Systems," in *IEEE Transactions on Industrial Informatics*, vol. 20, no. 2, pp. 2920–2931, Feb. 2024.